

YvO Proteomics Pipeline — Quality Control Summary

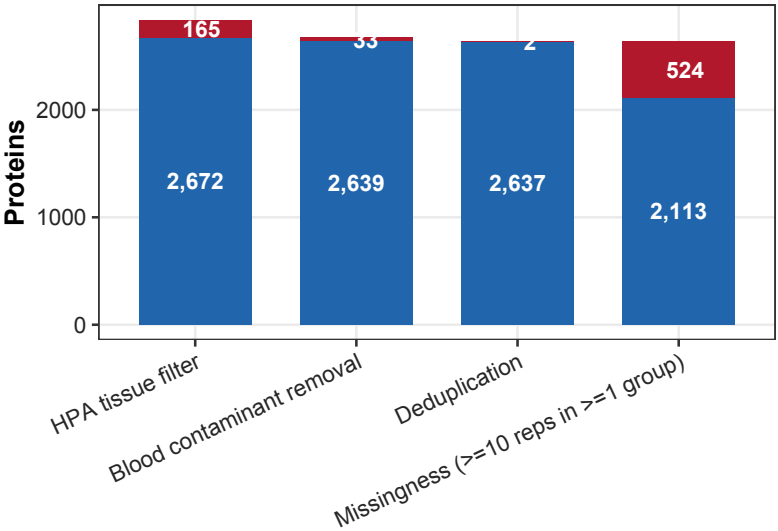
00_input -> 01_normalization (cyclic loess) -> 02_Imputation (missForest) | 2,113 proteins x 62 samples | 2026-04-18

A

Protein filter cascade

2,837 raw -> 2,113 retained

Removed Retained

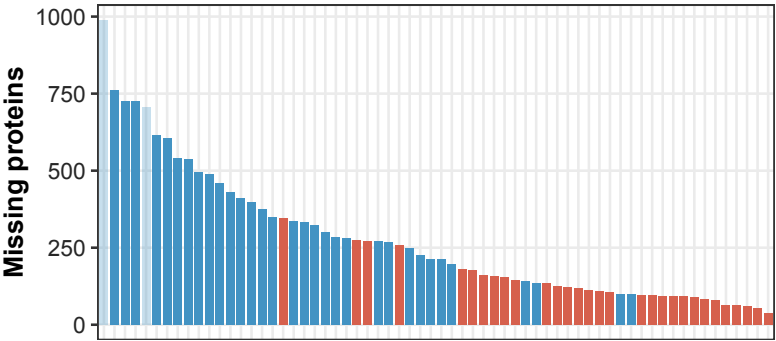


B

Per-sample missingness

64 samples | consensus outliers faded

Young_Pre Young_Post Old_Pre Old_Post

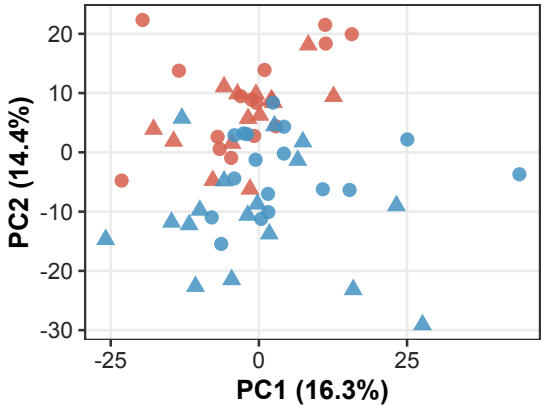


C

Pre-normalization PCA

PC1 16.3% | PC2 14.4%

Young Old Post Pre

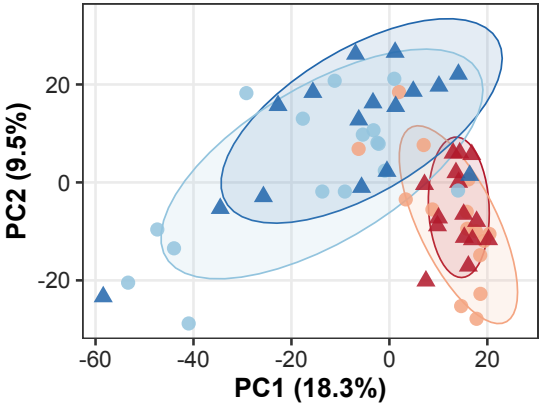


D

Post-normalization PCA

PC1 18.3% | PC2 9.5% | cyclic loess

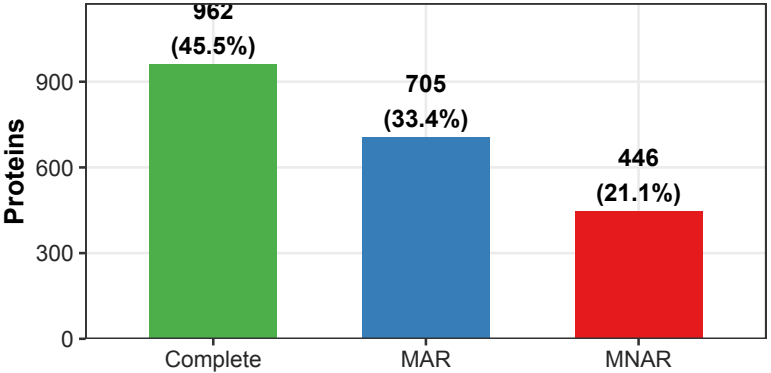
Young Pre Young Post Old Pre Old Post



E

Missingness classification

2,113 proteins | k-means (cluster means: 17.1 vs 19.6) classification



F

Imputation effect

115,173 observed | 15,833 imputed | OOB = 0.109

Observed Imputed (missForest)

